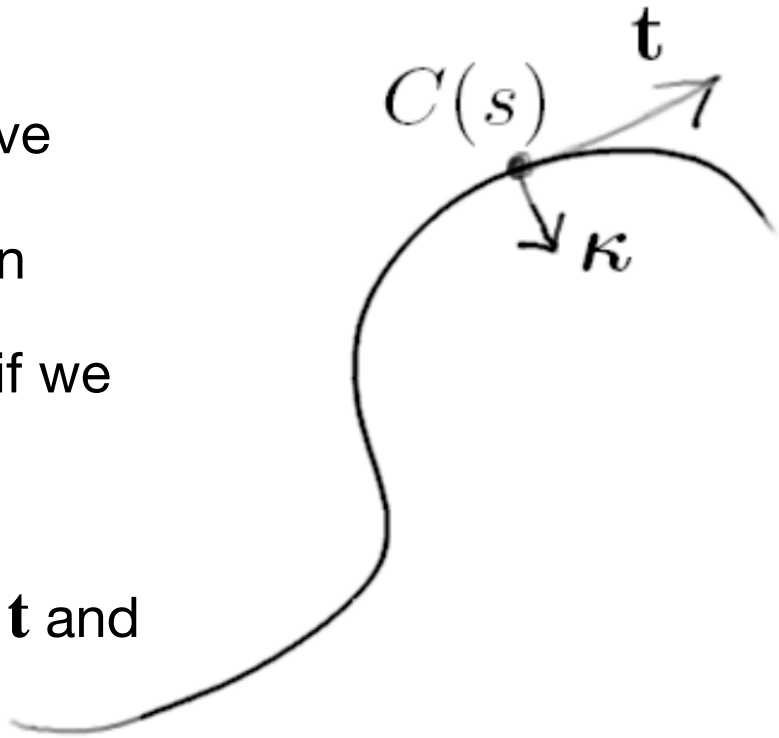
Principal Curvature

J. Andreas Bærentzen

Curvature

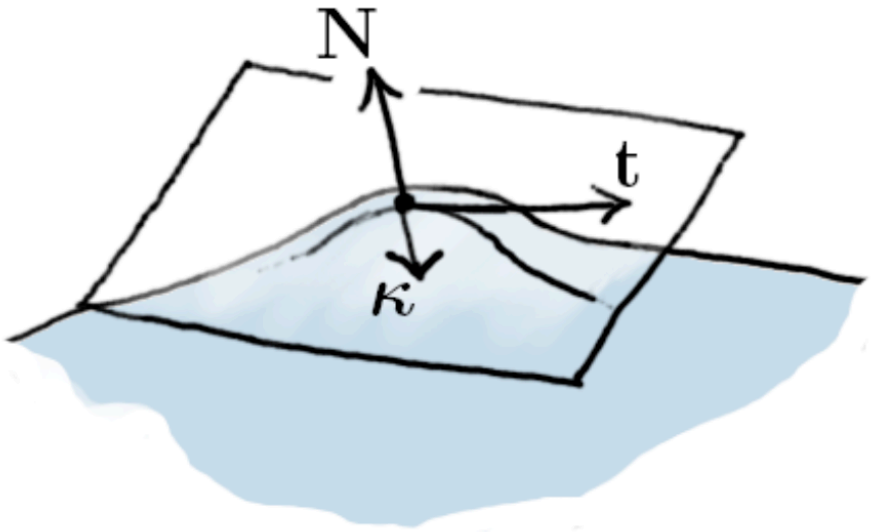
Is really only defined for plane curves

- We can understand
  - tangent, $\mathbf{t} = C'(s)$, as the velocity of the curve
  - The curvature, $\kappa = C''(s)$, as the acceleration
  - What do we know about these two vectors, if we move at constant speed?
  - Arc-length parametrisation means that $\|C'(s)\| = 1$ … what does that tell us about $\mathbf{t}$ and κ ?



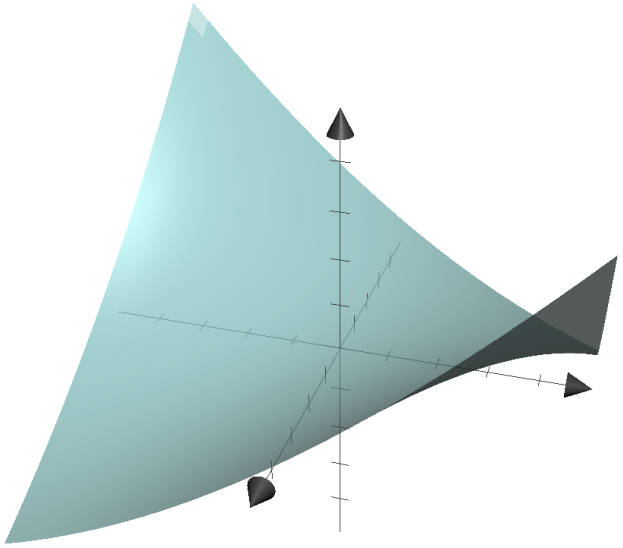
Notions of Curvature for Surfaces

- A *normal section* is a curve that is contained in the plane spanned by $\mathbf{N}$ and $\mathbf{t}$
- *Normal curvature, κ,* is the curvature of a normal section
- The directions of zero normal curvature are known as *asymptotic directions*
- The directions of minimum and maximum normal curvature are known as *principal*
- The principal directions are perpendicular to each other
- The principal curvatures have the
  - same sign at *elliptic points,*
  - opposite sign at *hyperbolic points, and*
  - one is zero at *parabolic points*



Principal Curvatures

Fitting a Quadratic Surface



- We fit a height field,

  $$h(u,v) = \frac{1}{2}\left(au^2 + 2buv + cv^2\right)$$

  to the vertices in the neighbourhood of vertex $\mathbf{p}_i$ obtaining the local representation

  $$\mathbf{x}(u,v) = \left[u, v, \frac{1}{2}\left(au^2 + 2buv + cv^2\right)\right]$$

Computing the Normal Curvature

By considering a curve C

We parameterise the normal section: $\mathbf{r}(s) = \mathbf{x}(u(s), v(s))$ where s is arc length

$$\mathbf{r}(s) = \left[u(s),\, v(s),\, \frac{1}{2}(au^2(s) + 2bu(s)v(s) + cv^2(s))\right],$$

Taking the first derivative:

$$\mathbf{r}'(s) = \left[u'(s),\, v'(s),\, au(s)u'(s) + b(u'(s)v(s) + u(s)v'(s)) + cv(s)v'(s)\right]$$

Computing the Curvature

By considering a curve C

$\left[u''(0),\, v''(0), 0\right] \parallel \left[u'(0),\, v'(0), 0\right]$ because

- $\mathbf{r}$ is a plane curve and
- $\mathbf{x}$ is a height map

By arc length parameterisation $\mathbf{r}'(s) \cdot \mathbf{r}''(s) = 0$, so $u''(0) = v''(0) = 0$

Thus, the second derivative is

$$\mathbf{r}''(0) = \left[0,\, 0,\, au'(0)u'(0) + 2bu'(0)v'(0) + cv'(0)v'(0))\right]$$

Computing the Curvature

By considering a curve C

Let $\xi = u'(0)$ and $\eta = v'(0)$ and note that $\|\mathbf{r}'(0)\| = 1,$ hence $\xi^2 + \eta^2 = 1$

Plug into the formula for $\mathbf{r}''(0)$:

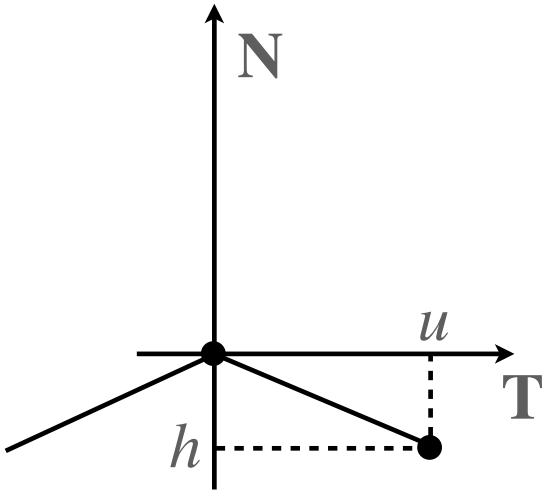
$$\kappa = \mathbf{r}''(0) = \left[0, 0, a\xi^2 + 2b\xi\eta + c\eta^2\right] = \left[0, 0, [\xi, \eta]\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} \xi \\ \eta \end{bmatrix}\right]$$

Written as a change to the normal, we have $\mathbf{S}\begin{bmatrix} \xi \\ \eta \end{bmatrix} = -\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} \xi \\ \eta \end{bmatrix}$

Project vertices onto tangent plane

Creating a local parametrization

- For each vertex,
  - Compute the normal $\mathbf{N}$ and the tangent plane $\mathbf{T}$
  - Project neighboring vertices using $(u_j, v_j) = \mathbf{T}^T(\mathbf{p}_j - \mathbf{p}_i)$
  - Get height values $h_j = \mathbf{N} \cdot (\mathbf{p}_j - \mathbf{p}_i)$
  - Find a, b, c such that $h_j = a\frac{u_j^2}{2} + bu_jv_j + c\frac{v_j^2}{2}$



Computing second order surface

- To fit the surface, we solve the following system in the least squares sense:

$$\begin{bmatrix} \vdots & \vdots & \vdots \\ \frac{u_j^2}{2} & u_j v_j & \frac{v_j^2}{2} \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \vdots \\ h_j \\ \vdots \end{bmatrix}$$

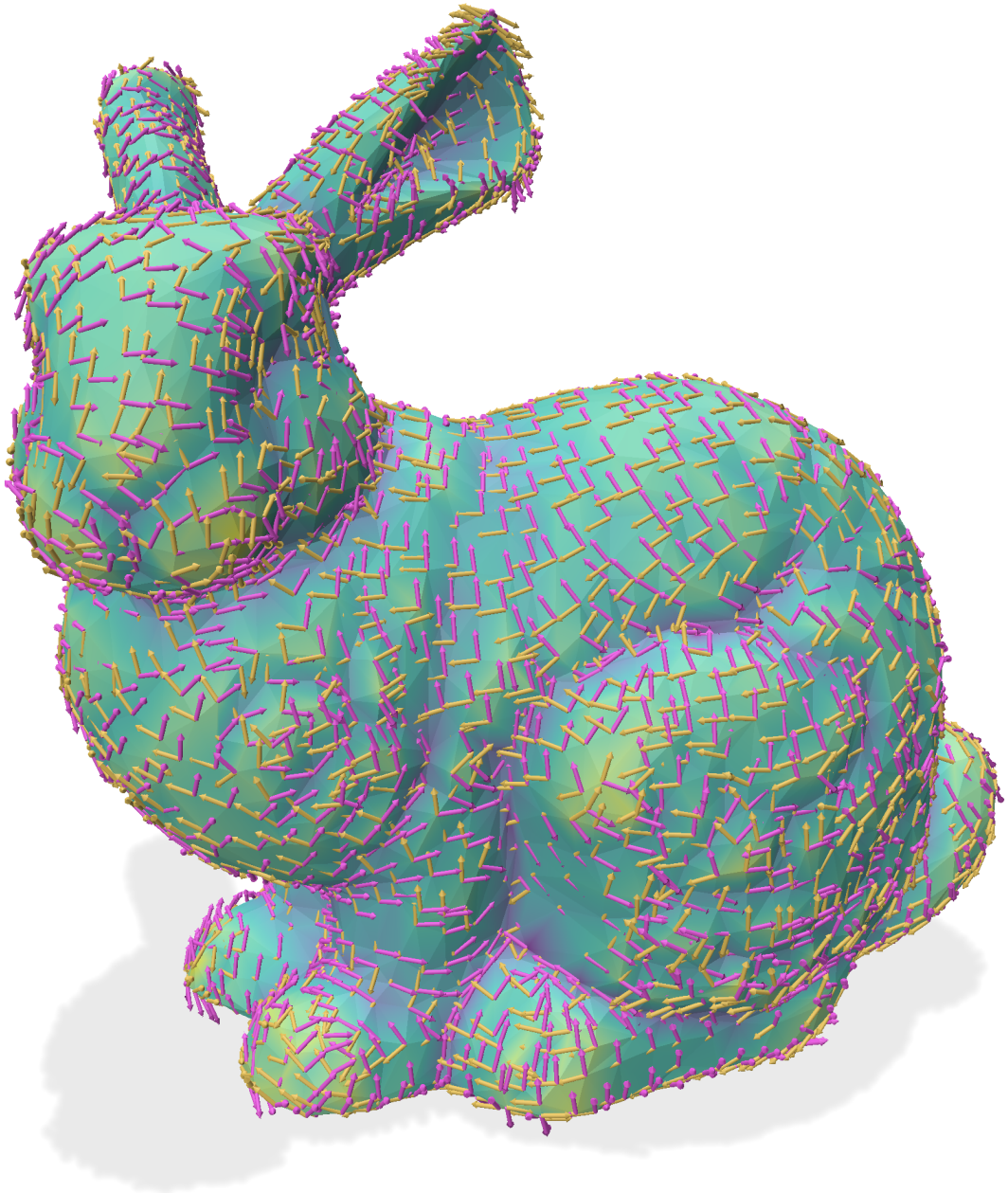
- Finally, the shape operator is

$$\mathbf{S} = -\begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

Principal curvature directions in 3D

- The principal curvatures are the eigenvalues of $\mathbf{S}$
- The principal directions are the eigenvectors of $\mathbf{S}$ (but in the tangent plane)
- To find principal directions in 3D, we can map the vectors, $\mathbf{d}_i$, using

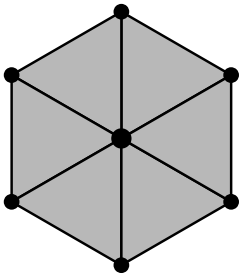
  $\mathbf{T}\mathbf{d}_i$ where $i \in \{0,1\}$ is one of the two principal directions



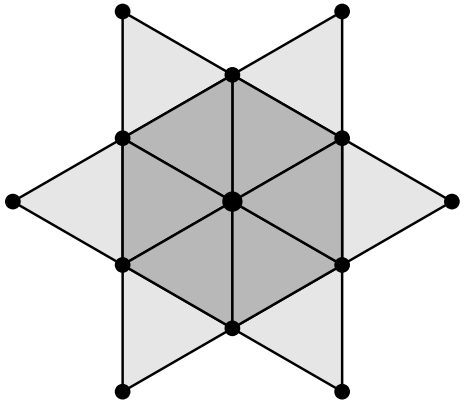
Tweaks

Can we make the shape operator more precise?

- There are three obvious changes to make
  - Instead of just using neighbours add flaps
  - Weight the vertices (Gaussian of distance)
  - Add linear terms to the quadratic

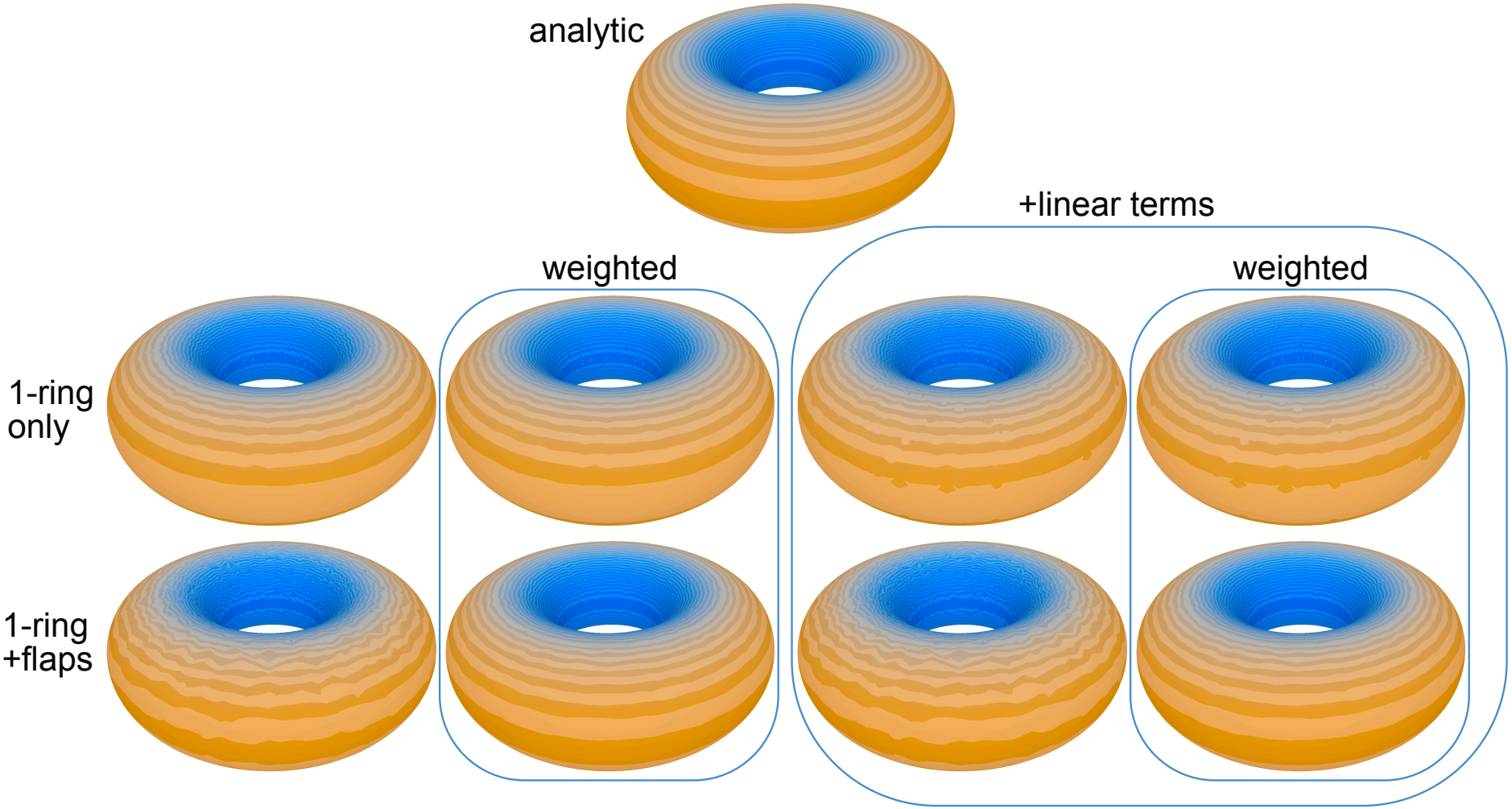


1-ring



1-ring + flaps

Comparison of Tweaks



Comparison of Tweaks

		Unweighted		Weighted	
Stencil	Polynomial	Mean	Max	Mean	Max
1-ring	quadratic	0.00771	0.123	**0.00518**	0.122
1-ring	quadratic + linear	0.01076	0.123	0.00831	0.122
1-ring + flaps	quadratic	0.01697	0.163	0.00679	0.090
1-ring + flaps	quadratic + linear	0.01701	0.156	0.00667	**0.082**

Curves on Surfaces

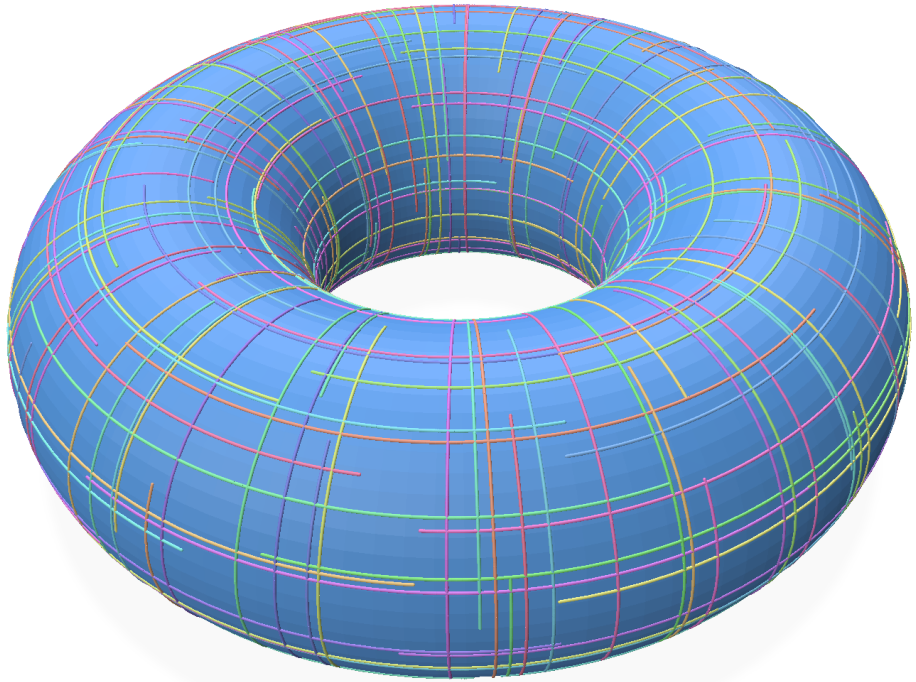
Lines of curvature

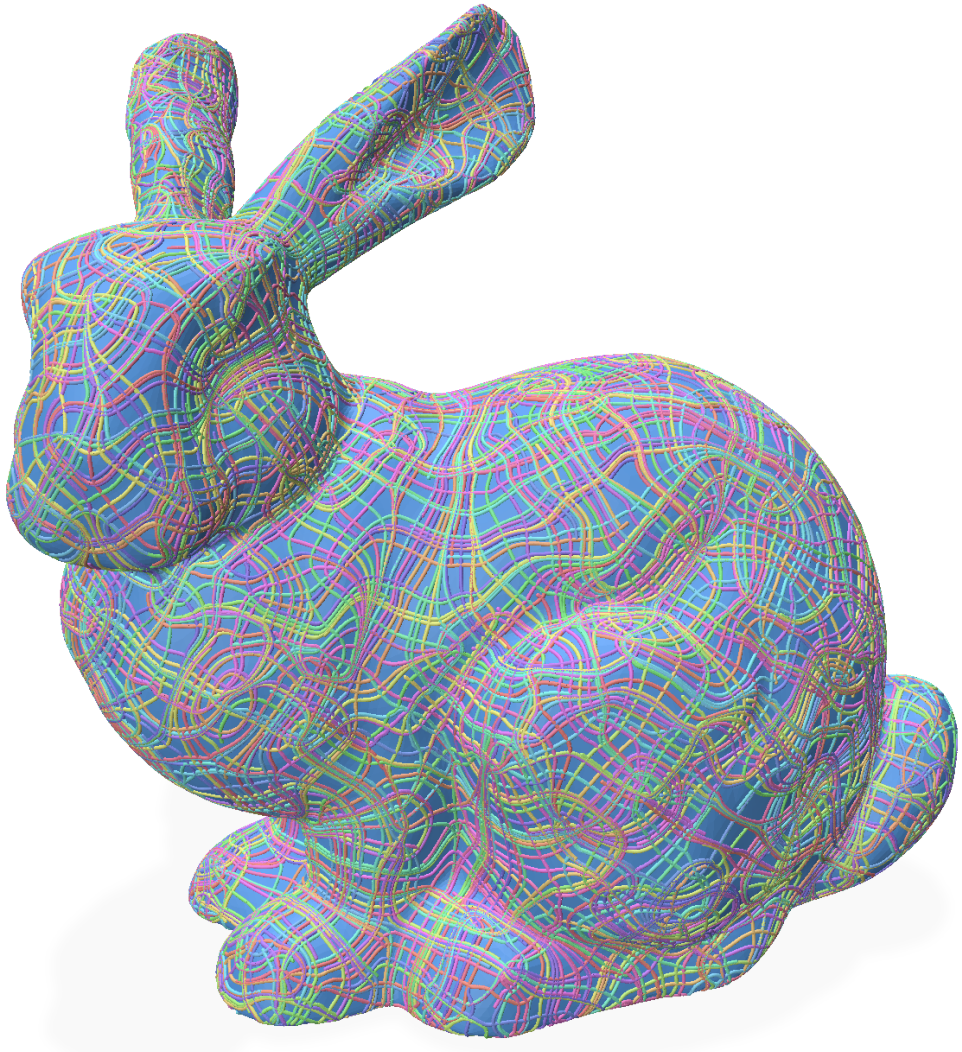
- Lines of curvature are formed by integrating principal curvature direction ($\mathbf{v}_1$ or $\mathbf{v}_2$)
- Starting at a point $\mathbf{p}$ we interpolate curvature direction to $\mathbf{p}$ and step in that direction
- Find the corresponding triangle, project into that triangle, compute barycentric, rinse and repeat!
- Note that
  - A line of curvature is always smooth away from umbilical points
  - Lines of curvature are not defined at umbilical points where $\kappa_{\min} = \kappa_{\max}$
  - On a compact surface with no boundaries, a line of curvature, which does not approach an umbilical point is a closed curve.
  - Along a line of curvature, it may change whether the corresponding curvature is numerically greatest or smallest

Curves on Surfaces

Lines of curvature

- Lines of curvature are formed by integrating principal curvature directions

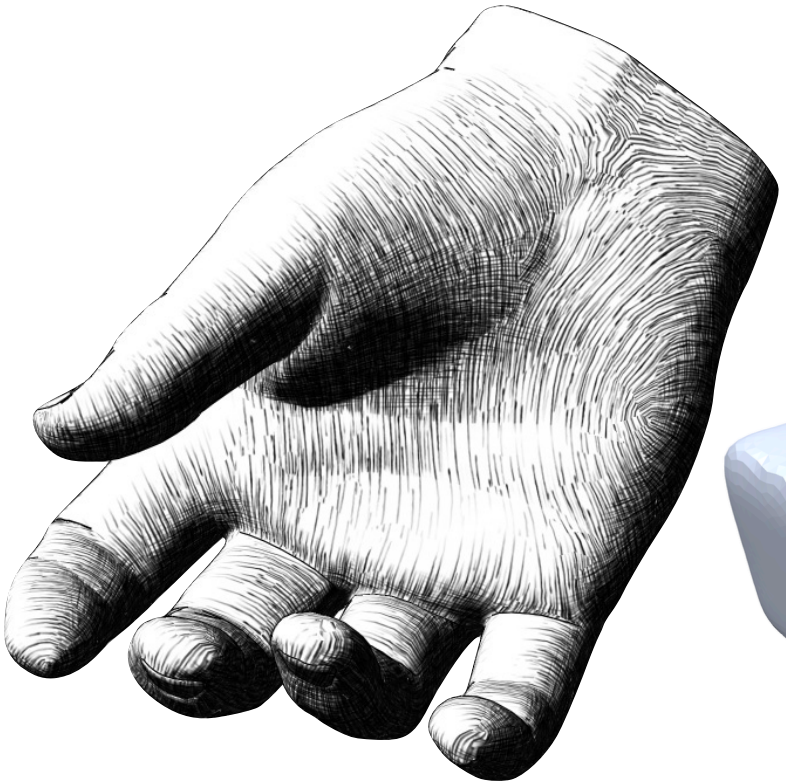




Asymptotic Lines

- Euler's formula states that the normal curvature is
  $\kappa_n(\theta) = \cos^2(\theta)\kappa_{\min} + \sin^2(\theta)\kappa_{\max}$
- We can find the asymptotic directions by solving for θ
  $\kappa_n(\theta) = 0$
- The corresponding asymptotic direction is now
  $\cos(\theta)\mathbf{v}_1 + \sin(\theta)\mathbf{v}_2$
- Note that asymptotic directions are defined only where $\kappa_{\min}\kappa_{\max} < 0$

Curvature Lines: Applications



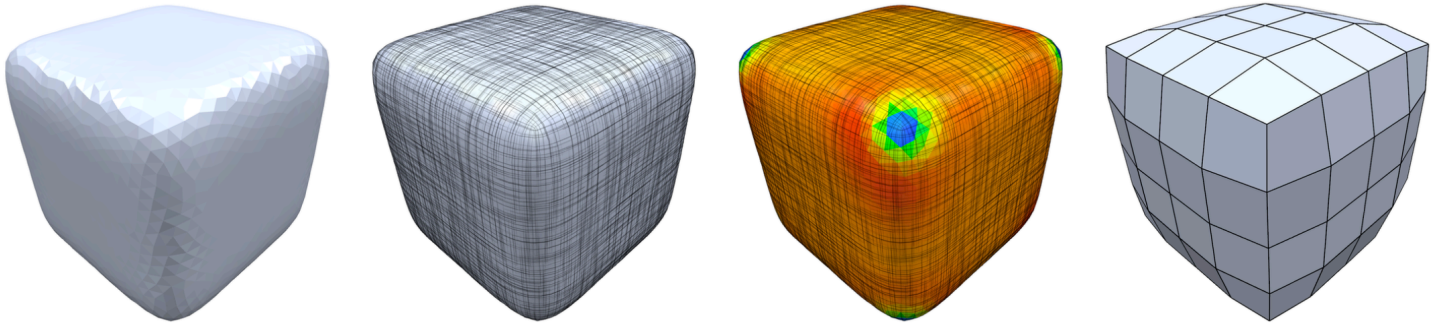
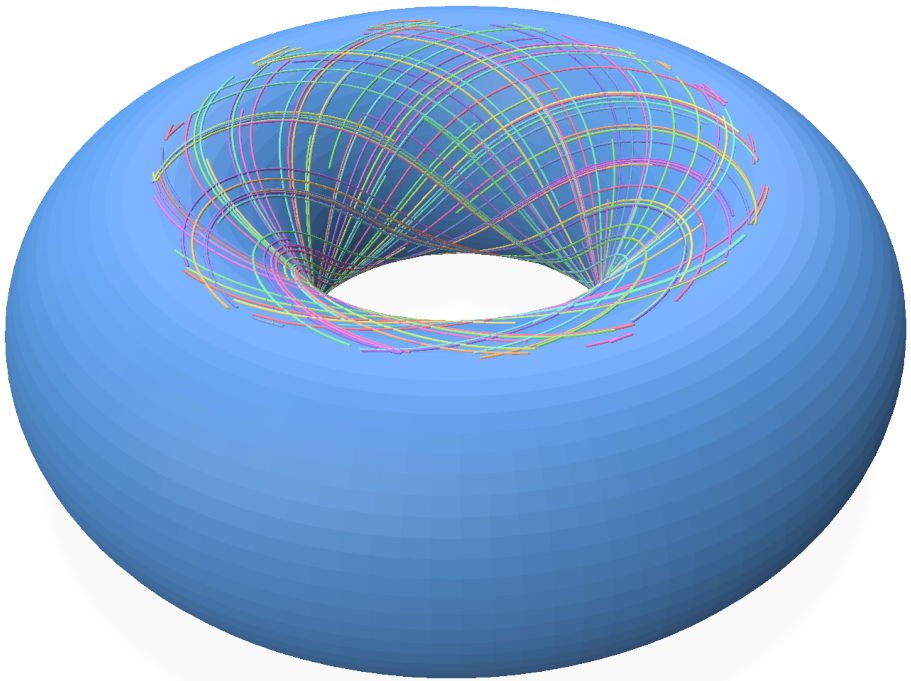


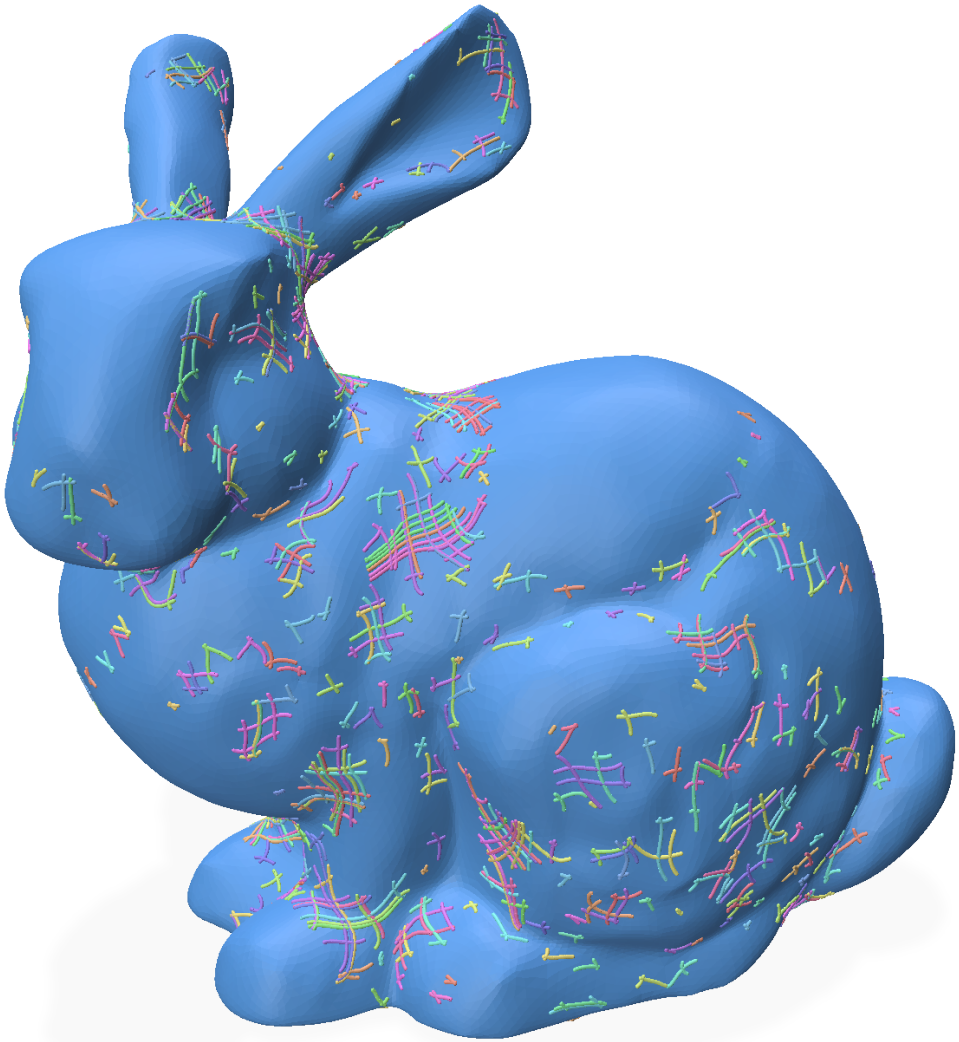
Figure 7: Prototype of a field guided method: Given an input triangle mesh (a) in the first step an orientation field (b) is computed which represents the local rotation of quad elements. In the second step a sizing field (c) is determined which specifies the sample density, which in this example is isotropic and close to uniform, with slight deviations color coded from blue to red. In the third step, a consistent quadmesh (d) is generated that closely reproduces both guiding fields.

Curves on Surfaces

Asymptotic lines

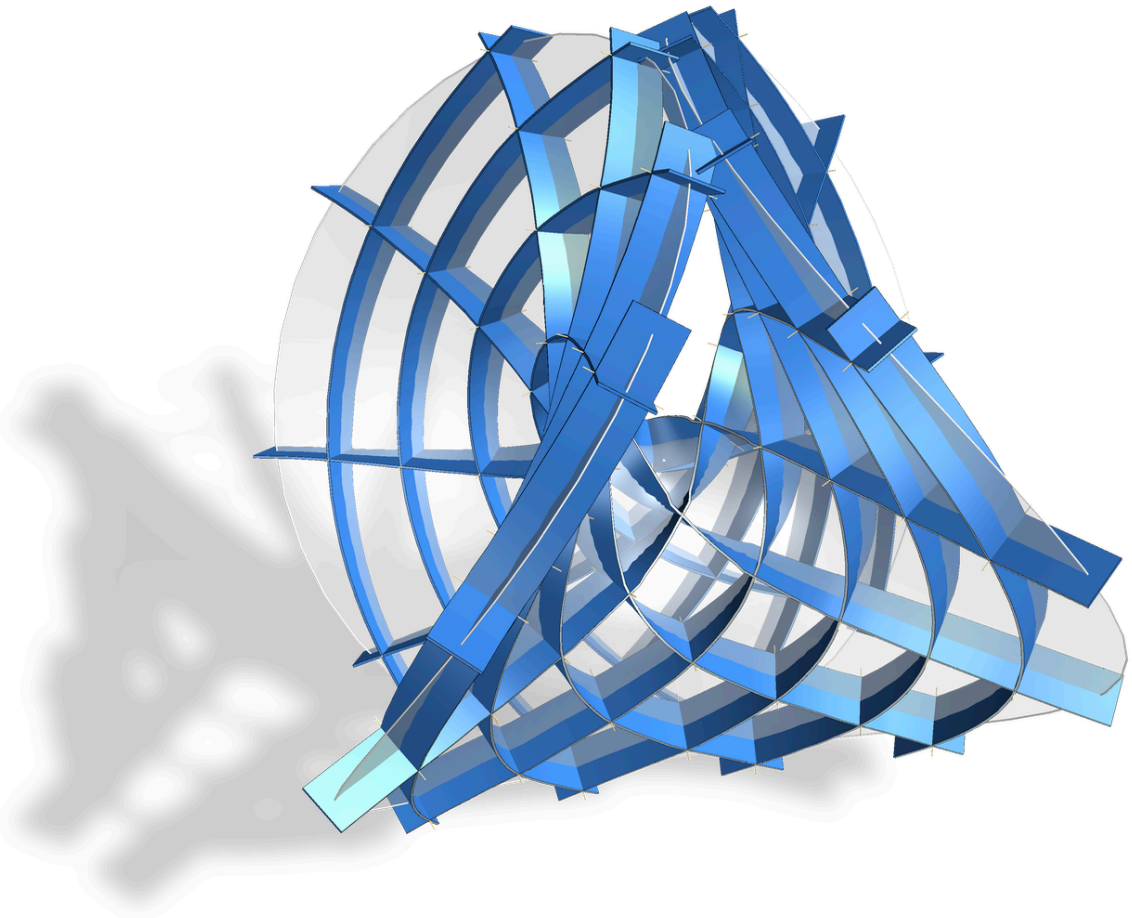
- Asymptotic lines are formed by integrating asymptotic directions





Asymptotic Nets

Courtesy of Quentin Becker

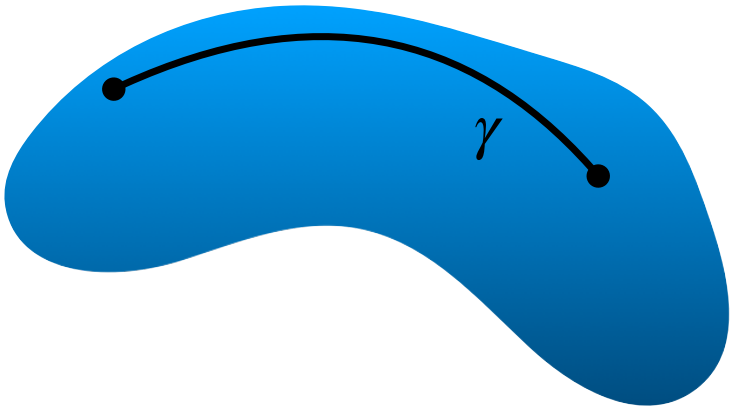


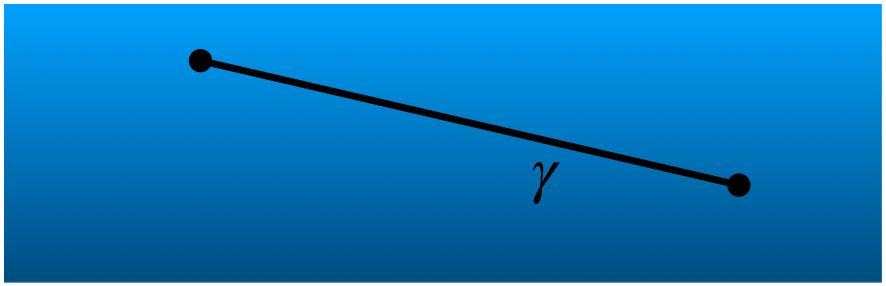
UTokyo

Curves on Surfaces

Geodesics

- Geodesic curvature is the curvature perpendicular to normal curvature
  - Geodesics are curves with zero geodesic curvature
  - Shortest path between any two points on a surface is a geodesic.





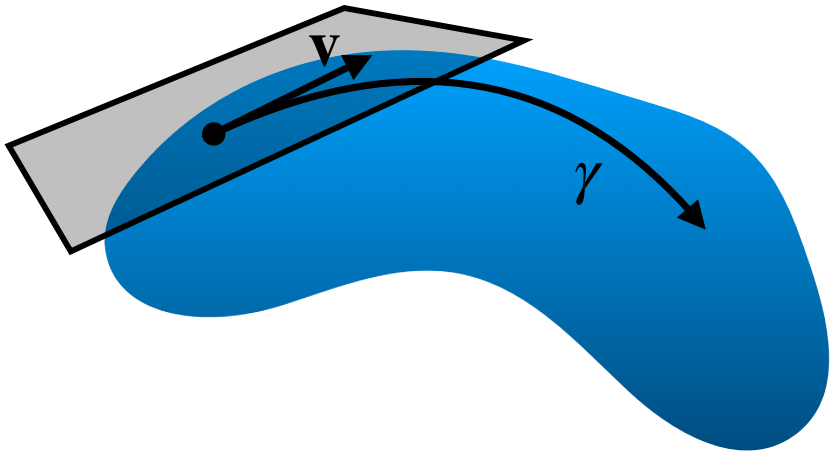
Bonus Slides

$\exp_R : \mathrm{T}_M \rightarrow M$

- The Riemannian exponential,

  $\exp_R(t\mathbf{v}) = \gamma(t)$, where $\|\mathbf{v}\| = 1$

  maps tangent vector $t\mathbf{v}$ to a geodesic, γ, of length t



The Discrete Exponential Map

A Map between a 2D domain and a 3D surface

- The discrete $\exp$ maps from a tangent plane to the surface
- The discrete $\log$ map is the inverse:
  - used to map a checkerboard texture from the tangent plane to the surface
- This provides a way to sample a surface consistently up to rotation
- Computation similar to Dijkstra's algorithm: [Melvær and Reimers. Geodesic polar coordinates on polygonal meshes. Computer Graphics Forum, 31(8), 2012]

